

Internship Report
Of
SPAM EMAIL CLASSIFIER

Submitted

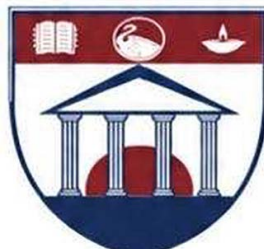
To

School of Computer Science and Engineering

IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement of degree of

B.Tech CSE



By

Roll Number of Student: 25SCS1003000673

Name of Student: AMITESH KUMAR

Batch: 2025-29

2026-27

CANDIDATE'S DECLARATION

I, AMITESH KUMAR do hereby declare that the internship report titled "**Spam Email Classifier**" has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in CSE. All the references have been quoted and I have not taken as such any material from any other source. I affirm to you that this report is my own and has not been submitted to any other institute for any degree/diploma requirement and further I shall be solely responsible for any kind of copyright violation in this regard.

Signature

Amitesh Kumar

Student Name: AMITESH KUMAR

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Topic: SPAM EMAIL CLASSIFIER

ACKNOWLEDGEMENT

I am using this opportunity to express my gratitude to everyone who supported me throughout the Internship Program. I am thankful for their aspiring guidance, invaluable constructive criticism and friendly advice during this work. I am sincerely grateful to them for sharing their truthful and illuminating views on a number of issues related to this work.

I express my gratitude to my parents for their blessings.

Signature: Amitesh Kumar

Student Name: Amitesh Kumar

Roll No.: 25scs1003000673

INTERNSHIP OFFER LETTER

Dear Amitesh Kumar ,

We are pleased to offer you a 1 Month Internship as Project Intern at Codec Technologies, a global platform dedicated to empowering learners and connecting diverse talent to create meaningful career opportunities worldwide. Codec Technologies delivers dedicated IT and business consultancy services across 27+ countries, empowering global innovation and strategic growth.

Internship Details:

- Designation : Artificial Intelligence Intern
- Location & Mode: Pan India, Hybrid
- Duration : 10/06/2026 to 10/07/2026
- Reporting to : Assigned Project Head(s)

This program provides an immersive experience, enabling you to gain industry-level knowledge and acquire valuable skills. Exceptional interns may also qualify for a scholarship award based on weekly performance reviews conducted by project supervisor.

If you accept this offer and the terms above, please complete your assigned projects and training tasks on our platform and via email.

Best regards,

Sincerely,

Dr. Anurag Shrivastava,
Codec Technologies India
NCS ID: E19E86-0116588288923
Date Of Issue - 10/06/2026



Dr. Anurag Shrivastava
Talent Acquisition Manager



Google for Education
Partner



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1. PROJECT DESCRIPTION

The project titled **Spam Email Classifier for Customer Service** focuses on developing a machine-learning based system that automatically identifies whether an incoming email is legitimate (ham) or unwanted/spam. Customer-service teams receive large volumes of messages every day, including promotional messages, phishing attempts, irrelevant advertisements and genuine customer requests. Manual filtering is time-consuming and can delay important responses.

The proposed classifier processes email subject and body text, cleans and normalizes the content, converts language into numerical features, trains supervised classification models, compares their performance and improves the selected model through parameter tuning and ensemble techniques. The resulting system can be integrated into a customer-service workflow to prioritize genuine requests and route suspicious messages for review.

Project Information	Details
Student	Amitesh Kumar
Roll Number	25scs1003000673
University	IILM University, Greater Noida
Organization	Codec Technologies
Internship	10/06/2026 – 10/07/2026
Project Title	Spam Email Classifier for Customer Service

The project demonstrates practical application of Natural Language Processing (NLP), machine learning, data preparation and model evaluation in a customer-service environment.

2. INTRODUCTION

Email remains an important communication channel for customers and organizations. Along with useful messages, inboxes may contain spam, unsolicited marketing, malicious links, scams and automated messages. If such messages are not filtered effectively, they consume employee time and can expose organizations to security and operational risks.

Spam classification is a text-classification problem. Each email is represented by its linguistic content and other measurable characteristics, and a trained model predicts a class such as spam or ham. A typical machine-learning pipeline includes data collection, cleaning, text preprocessing, feature extraction, model training, evaluation and deployment.

For customer service, the objective is not simply to delete unwanted messages. The system should support human agents by reducing noise while minimizing false negatives, because a genuine customer complaint or support request incorrectly classified as spam may have a significant business impact. Therefore, precision, recall, F1-score and confusion-matrix analysis are important in addition to accuracy.

This internship project applies standard NLP methods such as tokenization, lowercasing, stop-word handling and TF-IDF vectorization, followed by supervised classifiers such as Naive Bayes, Logistic Regression, Support Vector Machine and tree-based methods. Model comparison and ensemble learning are used to identify a robust solution.

3. PROFILE OF ORGANISATION – CODEC TECHNOLOGIES

Codec Technologies is considered in this report as the internship organization through which the project work was undertaken. The organization provides an environment for learning and applying software-development and technology concepts, including programming, data processing, analytics and artificial intelligence-related workflows.

During the internship, the project provides practical exposure to the lifecycle of a machine-learning application: understanding a business problem, preparing a dataset, selecting features, training algorithms, evaluating results and documenting the solution. The organization context helps connect academic concepts with an applied customer-service use case.

Learning and project environment: The internship work emphasizes structured problem solving, Python-based implementation, data handling, machine-learning experimentation and technical documentation. Such an environment is useful for developing the skills required to build deployable data-driven applications.

Relevance to the organization: Automated email classification can assist organizations that handle high volumes of customer communication. It can reduce repetitive manual screening, improve inbox organization and help service agents concentrate on legitimate customer interactions.

4. PROBLEM STATEMENT

Customer-service inboxes can contain a mixture of legitimate requests and unwanted emails. Manual identification of spam is inefficient when message volume increases. A reliable automated system is therefore required to classify incoming emails based on their textual content.

The major problem is to build a classifier that can distinguish spam from legitimate customer emails while controlling false classifications. The system must handle noisy text, variations in vocabulary, punctuation, capitalization and repeated promotional language. It should also be simple enough to integrate into a practical customer-service workflow.

The project addresses these issues by creating an end-to-end NLP and machine-learning pipeline and by comparing multiple classification algorithms before selecting an improved model.

Key challenges include inconsistent text, class imbalance, high-dimensional vocabulary, unseen words, overfitting and the need to balance false positives against false negatives.

5. PROJECT OBJECTIVES

- Collect or use a labeled email dataset containing spam and legitimate messages.
- Clean duplicate, empty and inconsistent records before model development.
- Pre-process email text to reduce noise and create consistent linguistic representations.
- Engineer useful textual and metadata-based features from subject and message content.
- Convert text into numerical vectors using TF-IDF and related representations.
- Train and compare multiple supervised machine-learning classifiers.
- Improve the selected model using hyperparameter tuning and suitable preprocessing.
- Explore ensemble modeling to improve robustness and generalization.
- Evaluate the classifier using accuracy, precision, recall, F1-score and a confusion matrix.
- Develop a solution concept that can support customer-service email triage.

6. SCOPE OF THE PROJECT

The scope covers binary classification of emails into spam and legitimate categories. The project includes data cleaning, NLP preprocessing, feature extraction, supervised model training, comparison, improvement and ensemble experimentation.

The solution can be extended to customer-service triage by adding confidence scores, priority classification, category routing and integration with an email server or help-desk platform. It can also be adapted for multilingual or domain-specific datasets after suitable retraining.

The current scope does not attempt to replace human security review, guarantee detection of every phishing campaign, or make irreversible decisions solely from model predictions. A production system should include monitoring, feedback collection, privacy controls and periodic retraining.

7. TECHNOLOGIES AND TOOLS USED

Technology / Tool	Purpose
Python	Core programming and model-development language
Pandas	Dataset loading, cleaning and manipulation
NumPy	Numerical operations
Scikit-learn	Vectorization, classification, evaluation and tuning
NLTK / spaCy	Natural-language preprocessing utilities
Matplotlib / Seaborn	Exploratory analysis and visualization
Jupyter Notebook	Experimentation and documentation
CSV / Excel	Dataset storage and inspection
Git	Version control and experiment tracking

Python is suitable because it provides a mature ecosystem for data science and NLP. Scikit-learn offers reliable implementations of common text-classification algorithms, while Pandas simplifies structured data preparation.

8. SYSTEM ARCHITECTURE

**Input Email → Data Validation → Data Cleaning → Text Pre-processing → Feature Engineering →
TF-IDF Vectorization → Train/Test Split → Model Training → Model Comparison →
Hyperparameter Tuning → Ensemble Model → Evaluation → Spam/Ham Prediction**

The architecture begins with an email containing a subject and body. Data validation checks for missing or invalid records. The cleaning module removes duplicates and irrelevant artifacts. Text preprocessing normalizes the language. Feature engineering creates additional representations, while TF-IDF transforms text into sparse numerical vectors.

The processed dataset is divided into training and testing subsets. Several models are trained on the training set and evaluated on unseen data. Hyperparameters are tuned for promising models. If appropriate, complementary classifiers can be combined into an ensemble. The final prediction is then supplied to the customer-service workflow.

9. DATA CLEANING MODULE

Data cleaning is an essential first stage because machine-learning performance depends strongly on data quality. Email datasets can contain duplicate messages, missing subjects, empty bodies, HTML markup, unusual encodings and irrelevant formatting.

Field / Issue	Cleaning Action
Missing subject	Replace with empty value or a neutral placeholder
Missing body	Remove record if no usable text remains
Duplicate emails	Detect and remove duplicate records
HTML tags	Strip markup where appropriate
URLs / email addresses	Normalize or replace with tokens
Extra whitespace	Collapse repeated spaces
Encoding artifacts	Normalize text encoding
Labels	Standardize to spam / ham

After cleaning, basic statistics should be recorded, including total records, spam count, ham count, missing values and duplicate count. This provides traceability and helps identify class imbalance before training.

10. FEATURE ENGINEERING MODULE

Feature engineering converts raw email content into informative signals. The most important textual feature is generally the word or n-gram representation of the message. Additional features can capture characteristics commonly associated with spam.

Feature	Description
Word n-grams	Sequences of one or more words
Character n-grams	Sub-word patterns useful for noisy/spoofed text
TF-IDF weights	Importance of terms relative to the corpus
Message length	Number of characters or tokens
URL count	Number of detected web links
Special-character count	Frequency of symbols such as ! or \$
Capitalization ratio	Approximate proportion of uppercase characters

Feature selection should be guided by validation results rather than assumptions. Excessive features can increase computation and overfitting, while insufficient features may reduce the model's ability to distinguish different spam patterns.

11. TEXT PRE-PROCESSING MODULE

Text preprocessing prepares subject and body content for vectorization. Typical operations include converting text to lowercase, removing unnecessary punctuation, tokenizing words, handling stop words and optionally applying stemming or lemmatization.

A practical pipeline may preserve certain information such as URLs, currency symbols or repeated punctuation through special tokens rather than deleting it completely, because these patterns can be informative for spam detection. The same preprocessing steps must be applied consistently to training and future emails.

Example: 'WIN a FREE OFFER!!! Visit <https://example.com>' can be normalized into a representation containing tokens such as 'win', 'free', 'offer' and a URL token. The precise preprocessing strategy should be validated experimentally because aggressive cleaning can remove useful spam indicators.

12. METHODOLOGY

The methodology follows a reproducible machine-learning workflow. First, the labeled dataset is inspected and cleaned. Second, text is preprocessed and transformed into numerical features. Third, the data is divided into training and testing subsets using stratification where appropriate. Candidate classifiers are trained on the training data. Finally, models are evaluated and improved using validation-based parameter tuning.

A representative split is 80% training and 20% testing, although the exact split can be adjusted based on dataset size. Cross-validation on the training portion can be used to select hyperparameters without leaking information from the final test set.

12.1 TEXT VECTORIZATION

Machine-learning algorithms require numerical input. TF-IDF (Term Frequency–Inverse Document Frequency) is a strong baseline for spam classification because it emphasizes terms that are frequent in an individual email but less common across the overall corpus.

For a term t in document d , TF-IDF can be represented conceptually as $TF(t,d) \times IDF(t)$, where IDF reduces the influence of terms occurring in many documents. Word unigrams and bigrams can capture common phrases, while character n -grams can help with obfuscated words and spelling variations.

The vectorizer should be fitted only on training data and then used to transform validation and test data. This prevents information leakage from the evaluation set.

12.2 MODEL TRAINING AND COMPARISON

Several supervised algorithms can be evaluated. Multinomial Naive Bayes is computationally efficient for sparse text features. Logistic Regression provides a strong linear baseline and interpretable feature weights. Linear Support Vector Machine often performs well on high-dimensional text. Tree-based models can be considered for engineered metadata features.

Model	Strength	Consideration
Multinomial Naive Bayes	Fast and effective for word counts/TF-IDF	Assumes conditional independence
Logistic Regression	Strong linear text baseline	Needs regularization tuning
Linear SVM	Effective in high-dimensional sparse spaces	Probability estimates require extra calibration
Random Forest	Useful with mixed engineered features	Can be less efficient for huge sparse vocabularies
Voting / Stacking Ensemble	Combines complementary models	More complex to validate and deploy

Evaluation should include accuracy, precision, recall and F1-score. For customer service, recall for spam and precision for legitimate mail should both be monitored because incorrectly filtering a genuine customer message can be costly.

12.3 MODEL IMPROVEMENT

Model improvement begins with error analysis. Misclassified messages should be inspected to identify patterns such as promotional language, short messages, unusual spellings, HTML-heavy content or legitimate emails containing words frequently seen in spam.

Hyperparameter tuning can adjust regularization strength, n-gram ranges, minimum document frequency and maximum vocabulary size. Class weighting can be explored if the dataset is imbalanced. Threshold tuning may also be useful when the business prefers higher recall or higher precision.

Improvement must be measured on validation data and confirmed on a held-out test set. Repeated experimentation on the test set can lead to optimistic results, so the test set should remain isolated until the final evaluation.

12.4 ENSEMBLE MODELING

Ensemble learning combines predictions from multiple models. A simple soft-voting or hard-voting approach can combine complementary classifiers. Stacking can use predictions from base learners as inputs to a meta-classifier.

For example, a linear SVM may capture discriminative word patterns while a Naive Bayes model may respond strongly to characteristic spam vocabulary. Combining models can improve robustness when their errors are not perfectly correlated.

The ensemble should only be adopted if it produces a meaningful improvement on validation data and does not create unnecessary deployment complexity. In a customer-service environment, latency, interpretability and maintenance are also important considerations.

13. EXPECTED OUTCOMES

Outcome	Expected Benefit
Automated spam detection	Reduces manual inbox screening
Improved email prioritization	Helps agents focus on genuine customer requests
Consistent classification	Applies the same rules/model to large volumes
Reduced operational noise	Removes or flags repetitive unwanted messages
Measurable performance	Provides precision, recall, F1 and confusion matrix
Extensible architecture	Can support future routing and priority models
Learning outcome	Demonstrates practical NLP and ML skills

The expected technical outcome is a validated text-classification pipeline capable of distinguishing spam and legitimate messages with strong performance on unseen data. Exact performance depends on dataset quality, domain, class balance and preprocessing choices; therefore, numerical results should be reported only after running the final implementation on the selected dataset.

14. CONCLUSION

The Spam Email Classifier for Customer Service demonstrates how NLP and supervised machine learning can be applied to an everyday business problem. The project brings together data cleaning, text preprocessing, feature engineering, TF-IDF vectorization, model training, comparison, improvement and ensemble learning in a single workflow.

A reliable classifier can reduce unwanted email noise and help customer-service teams concentrate on legitimate requests. The project also highlights the importance of evaluating more than accuracy: precision, recall, F1-score, confusion matrices and error analysis are necessary to understand the real behavior of a classification system.

For future development, the system can be extended with multilingual support, phishing-specific detection, real-time email integration, confidence-based human review, feedback-driven retraining and monitoring for changes in spam patterns. With appropriate privacy and security controls, the classifier can become a useful component of a broader customer-service automation platform.

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CERTIFICATE OF INTERNSHIP

This certificate is awarded to
Amitesh Kumar

From Codec Technologies Pvt. Ltd.

In recognition of his/her efforts and achievements in completing the 1 Month AICTE & ICAC

Approved internship program as

Artificial Intelligence Intern

Conducted from 10/06/2026 to 10/07/2026



Google for Education
Partner



AICTE ID - CORPORATE6759d549ce59e1733940553

A handwritten signature in black ink, belonging to Dr. Anurag Shrivastava.

Dr. Anurag Shrivastava
Program Manager